

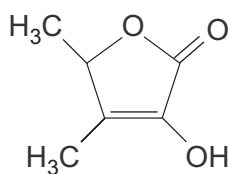
SECTION A

Answer all questions in the spaces provided.

1. State what is seen, if anything, when aqueous phenol is added to aqueous solutions of the following reagents. [2]

Reagent	Observation
iron(III) chloride	
sodium hydroxide	

2. Sotolone is one of the compounds that is responsible for the smell of raspberries.



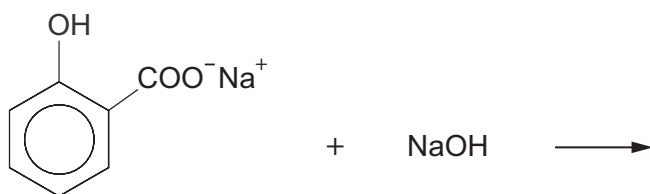
sotolone

Give the **empirical** formula of this compound.

[1]

.....

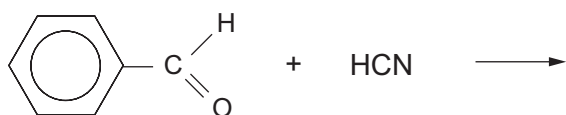
3. Complete the equation for the reaction of sodium 2-hydroxybenzoate with soda lime (NaOH), which, on strong heating, can give phenol as one of the products. [1]



SECTION A

Answer **all** questions in the spaces provided.

1. (a) Complete the equation below, which shows the reaction of benzaldehyde with hydrogen cyanide. [1]



- (b) Give the **formula** of the nucleophile that takes part in the reaction shown in part (a). [1]

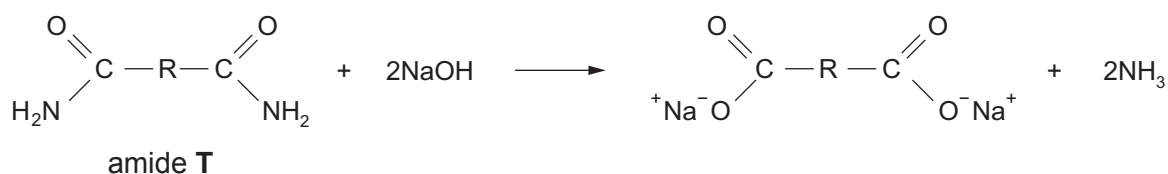
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2. Decarboxylation occurs when a carboxylic acid is heated with sodalime.

State the name of the carboxylic acid that will produce propane when strongly heated with sodalime. [1]

.....

3. Ammonia is produced when an amide is heated with aqueous sodium hydroxide.



In an experiment 3.90 g of the aliphatic amide **T** gave 0.060 mol of ammonia.

Use this information to find the formula of the hydrocarbon group R, present in the amide **T**. [2]

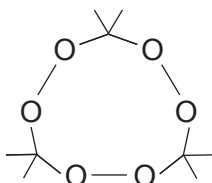
Formula



SECTION A

Answer **all** questions in the spaces provided.

1. The skeletal formula of the compound TATP is shown below.



Give the **molecular** formula of this compound.

[1]

.....

2. Alkanes are produced when carboxylic acids are strongly heated with sodalime.

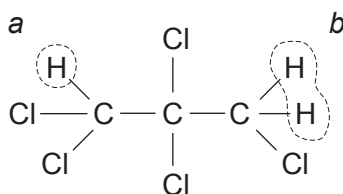
State the name of a carboxylic acid that will produce propane, when heated in this way.

[1]

.....

3. Complete the table below, which describes the ^1H NMR spectrum of 1,1,2,2,3-pentachloropropane.

[2]



Proton(s)	Splitting pattern	Relative peak area ratio
<i>a</i>		
<i>b</i>		



SECTION A

Answer **all** questions.

1. A hydrocarbon is either cyclopentane or pent-1-ene.

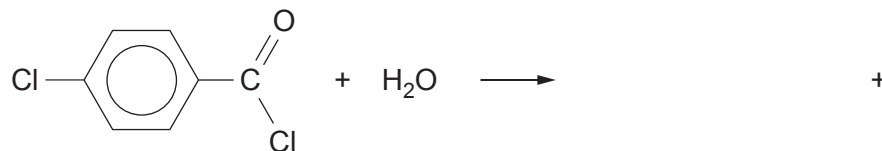
Use the **Data Booklet** to choose a characteristic infrared absorption that will positively identify the compound as pent-1-ene. [1]

.....
.....

2. Give the **name** of a compound of formula C_4H_8 that will decolourise acidified potassium manganate(VII). [1]

.....

3. Complete the equation for the reaction shown below. [1]



4. Give the structure of a compound of molecular formula $C_4H_8Br_2$ whose low resolution 1H NMR spectrum consists of three peaks. [1]



SECTION AAnswer **all** questions.

1. Give the structure of an unsaturated aldehyde of molecular formula C_4H_6O . [1]
2. State a group that will give a positive triiodomethane (iodoform) test and give the observation for a positive result. [2]
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3. (a) 1,2-Diaminoethane reacts as a base. [1]
- Explain how this compound acts as a base.
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-
- (b) Give the structure of the organic compound formed when 1 mole of 1,2-diaminoethane reacts with 2 moles of ethanoyl chloride. [1]

